

Deriving the Metropolis-Hastings Transition Kernel from First Principles

Monte Carlo Statistical Methods

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1 Metropolis-Hastings as Inverse Markov Chain Design

In this section we derive the Metropolis–Hastings (MH) algorithm from first principles. The goal is not merely to present the final formula, but to understand *why* each ingredient has the form that it does. Everything will flow from a single viewpoint:

Constructing MH is an *inverse problem*: we know the desired stationary distribution and must design a Markov chain that has it.

We work on a general state space $\mathcal{X}$, possibly $\mathbb{R}^d$, and assume we know an unnormalized version of the target density.

1.1 The Problem: Sampling by Designing a Markov Chain

We want to sample from a target density

$$f_X(x) = \frac{1}{Z} \tilde{f}(x), \quad Z = \int \tilde{f}(x) dx \text{ unknown or intractable.}$$

Direct sampling (inverse CDF, transforms, classical accept–reject) often fails when $\tilde{f}$ is high-dimensional or irregular. The Metropolis–Hastings idea is to give up on *independent* samples, and instead generate a sequence

$$X_0, X_1, X_2, \dots$$

from a Markov chain that eventually *settles* into the distribution f_X .

So we seek a transition kernel $T(x, y)$ satisfying:

- $T(x, y) \geq 0$, and $\int T(x, y) dy = 1$ (valid Markov kernel).
- f_X is stationary:

$$f_X(y) = \int f_X(x) T(x, y) dx.$$

This is the *inverse* of the classical problem in Markov chain theory: normally T is given and we solve for the stationary distribution; here f_X is given and we must construct T .

1.2 Stationarity, Flow, and Detailed Balance

To understand how to construct such a kernel, it helps to visualize how probability moves in a chain.

Define the *probability flow*:

$$J(x, y) := f_X(x) T(x, y).$$

If the chain is at equilibrium, $J(x, y)$ is the amount of probability mass moving from x to y in a single step.

Stationarity says:

$$f_X(y) = \int J(x, y) dx.$$

Furthermore, the fact that $T(x, \cdot)$ integrates to 1 means

$$\int J(x, y) dy = f_X(x).$$

So probability flow *out of* x must match the stationary mass at x .

A far simpler condition—one that automatically implies stationarity—is the symmetry of flows:

$$J(x, y) = J(y, x).$$

This is *detailed balance*:

$$f_X(x) T(x, y) = f_X(y) T(y, x).$$

It states that across every “edge” (x, y) there is no net drift of probability. Equilibrium is achieved not by complicated global cancellations, but by simple pairwise symmetry.

Thus a practical strategy is:

Design T to satisfy detailed balance with respect to f_X . Then stationarity is guaranteed.

The remaining challenge is: *how to actually build such a T ?*

2 Introducing Proposals: Why We Need a Second Distribution

We face a practical issue: even if we know f_X , we cannot in general sample from it directly or even manipulate it easily. So we introduce a second density $q(y | x)$ that is easy to sample from.

But this raises a conceptual question:

Why bring in a proposal density at all? Why not construct T directly?

Because:

- We need T to be a probability density in y for each x .
- We need T to satisfy detailed balance.
- We need T to be easy to simulate.

Directly solving those three requirements at once is extremely difficult.

Introducing $q(y | x)$ solves the *simulation* problem immediately: we will build T out of q plus some corrections, so that T is easy to simulate from by a simple two-step procedure.

This mirrors what accept–reject sampling does for a single variable: propose first, then selectively accept.

Here we will propose Y from $q(\cdot | x)$ and adjust this proposal with an acceptance probability.

Thus the guiding thought is:

If I can simulate from $q(y | x)$, maybe I can “distort” it using acceptance/rejection to produce the correct stationary flow.

3 Raw Capacities: The First Clue About Acceptance

Suppose the chain *were already* distributed as f_X . If we always accepted proposals, then the raw probability flow from x to y would be

$$C_{x \rightarrow y}^{\text{raw}} := f_X(x) q(y | x).$$

Similarly, the raw flow from y to x is

$$C_{y \rightarrow x}^{\text{raw}} := f_X(y) q(x | y).$$

These flows reflect exactly how much mass would travel if no correction were applied. However, they are *not* symmetric in general, so they do not satisfy detailed balance.

This suggests a very natural idea:

Maybe we can correct the raw proposal flows by selectively accepting or rejecting them so that the *final* flows become symmetric.

This is where the acceptance probability enters the story.

4 Thinning Proposals: Introducing the Acceptance Function

To control the final flow, we attach an acceptance probability $a(x, y) \in [0, 1]$ to moves from x to y . We then define

$$T(x, y) = q(y | x) a(x, y) \quad (y \neq x).$$

This expresses the transition as a two-step sampling mechanism:

1. Propose $Y \sim q(\cdot | x)$.
2. Accept it and move to Y with probability $a(x, Y)$.

Rejected proposals cause the chain to stay at x .

Now the *actual* flow becomes

$$J(x, y) = f_X(x) q(y | x) a(x, y).$$

This is a tunable flow: it is bounded above by the raw capacity $f_X(x) q(y | x)$, and we can choose $a(x, y)$ to achieve symmetry.

We now arrive at the precise mathematical task:

Choose $a(x, y)$ such that the flows $J(x, y)$ are symmetric and do not exceed the raw capacities.

5 Symmetric Flows: Solving for $a(x, y)$

Detailed balance requires

$$f_X(x) q(y | x) a(x, y) = f_X(y) q(x | y) a(y, x).$$

We have two unknowns $a(x, y)$ and $a(y, x)$. But we also know that both must lie in $[0, 1]$.

A clean and painless way to enforce symmetry is to define the *symmetric flow* on the pair (x, y) as

$$J(x, y) := \min\{f_X(x)q(y | x), f_X(y)q(x | y)\}.$$

This choice has an extremely intuitive meaning:

Send as much symmetric probability flow as possible along the edge (x, y) , but do not exceed either direction's raw capacity.

Now set

$$f_X(x)q(y | x)a(x, y) = \min\{f_X(x)q(y | x), f_X(y)q(x | y)\}.$$

Let $A = f_X(x)q(y | x)$ and $B = f_X(y)q(x | y)$. Then

$$a(x, y) = \frac{\min\{A, B\}}{A}.$$

So in the two possible cases:

- If $A \leq B$:

$$a(x, y) = 1.$$

Interpretation: forward raw flow is the bottleneck; accept all proposals in the forward direction.

- If $A > B$:

$$a(x, y) = \frac{B}{A} = \frac{f_X(y)q(x | y)}{f_X(x)q(y | x)}.$$

Interpretation: backward raw flow is the bottleneck; scale down the forward flow to match it.

Combining into a single expression gives the classic Metropolis–Hastings acceptance probability:

$$a(x, y) = \min\left(1, \frac{f_X(y)q(x | y)}{f_X(x)q(y | x)}\right).$$

We now have all off-diagonal entries of the kernel:

$$T(x, y) = q(y | x)a(x, y), \quad y \neq x.$$

To complete the construction, we must find $T(x, x)$.

6 Deriving the Diagonal Term $T(x, x)$

Now we compute

$$T(x, x) = \mathbb{P}(X_{n+1} = x | X_n = x)$$

by enumerating all ways the chain can stay at x after one MH step.

6.1 Propose x and accept

Contribution:

$$q(x | x) a(x, x).$$

6.2 Propose x and reject

Contribution:

$$q(x \mid x) [1 - a(x, x)].$$

Together, these two sum to $q(x \mid x)$.

6.3 Propose some $z \neq x$ and reject

Contribution:

$$\int_{z \neq x} q(z \mid x) [1 - a(x, z)] dz.$$

6.4 Sum all contributions

$$T(x, x) = q(x \mid x) + \int_{z \neq x} q(z \mid x) [1 - a(x, z)] dz.$$

Since $\int q(z \mid x) dz = 1$, we have

$$T(x, x) = 1 - \int_{z \neq x} q(z \mid x) a(x, z) dz.$$

Because $z = x$ contributes $q(x \mid x)a(x, x)$ to the integral, we can write the final form uniformly as

$$T(x, x) = 1 - \int q(z \mid x) a(x, z) dz.$$

7 Final Unified Expression

Introduce the Dirac delta (or Kronecker delta on discrete spaces):

$$\delta_{x,y} = \begin{cases} 1, & y = x, \\ 0, & y \neq x. \end{cases}$$

Then the Metropolis–Hastings transition kernel can be written in a single compact formula:

$$T(x, y) = a(x, y) q(y \mid x) + \left(1 - \int a(x, z) q(z \mid x) dz\right) \delta_{x,y}.$$

Interpretation:

- The term $a(x, y)q(y \mid x)$ corresponds to a proposed and accepted move to y .
- The scalar

$$1 - \int a(x, z) q(z \mid x) dz$$

is the total probability of rejecting a proposal, which forces the chain to stay at x .

- Detailed balance holds because we constructed $a(x, y)$ to symmetrize flows.

Thus, starting from the inverse Markov chain design problem and proceeding through proposal, capacity, and flow symmetry arguments, we obtain the Metropolis–Hastings transition kernel in full.

8 On Convergence: Stationarity is Not Enough

The construction above guarantees that the Metropolis–Hastings kernel $T(x, y)$ satisfies the detailed balance equation

$$f_X(x) T(x, y) = f_X(y) T(y, x),$$

and therefore f_X is a stationary distribution of the chain.

However, *stationarity alone is not sufficient* to justify using

$$X_0, X_1, X_2, \dots$$

as a sequential sampling method from f_X . To guarantee that the distribution of X_n converges to f_X for any reasonable initial state X_0 , the Markov chain must in addition be:

- **Irreducible:** every region of $\mathcal{X}$ with positive f_X -mass must be reachable (possibly in multiple steps) with positive probability;
- **Aperiodic:** the chain must not cycle deterministically between disjoint subsets;
- **Positive recurrent:** the expected return time to any region of positive measure must be finite.

These three properties, together with stationarity, ensure *ergodicity*:

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n \in A \mid X_0 = x) = \int_A f_X(y) dy \quad \text{for all measurable sets } A.$$

The Metropolis–Hastings algorithm satisfies these conditions under mild and transparent assumptions on the proposal kernel $q(y \mid x)$ (for example, that q has positive density in a neighborhood of each point). Those proofs are separate and are not part of the flow-symmetry construction above.

Thus, detailed balance shows that f_X is a stationary distribution of the MH kernel, and irreducibility, aperiodicity, and positive recurrence ensure that the MH chain indeed *converges* to f_X and can be safely used for sequential sampling.